

- 3 (a) State one similarity and one difference between an electric force and a gravitational force.

Similarity

.....

Difference

..... [2]

- (b) In a Rutherford's gold scattering experiment, an alpha particle with a kinetic energy of 20 keV approaches a stationary gold nucleus ($^{197}_{79}\text{Au}$) from a large distance away. It reaches a distance of closest approach $d_{\min}$ before being deflected 180° .

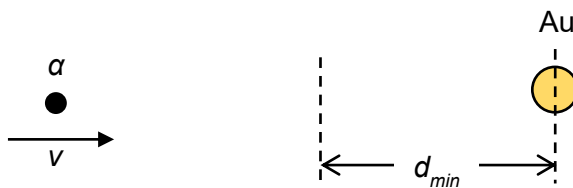


Fig. 3.1

- (i) Calculate the charge on the gold nucleus and the alpha particle.

charge of gold nucleus = C

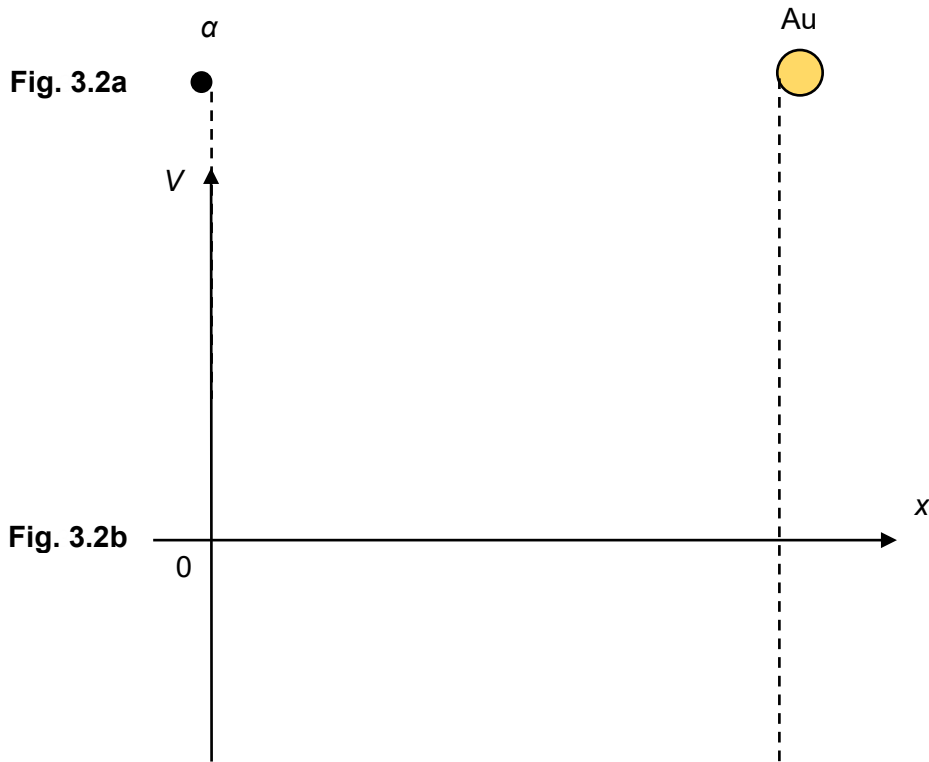
charge of alpha particle = C [2]

(ii) Calculate $d_{\min}$.

$d_{\min} = \dots\dots\dots \text{ m [3]}$

(iii) Fig. 3.2a shows the alpha particle at its distance of closest approach to the gold nucleus.

Sketch on Fig. 3.2b the variation with distance x of the resultant electrical potential V due to both the alpha particle and the gold nucleus.



[2]